sits: Satellite Image Time Series Analysis on Earth Observation Data Cubes

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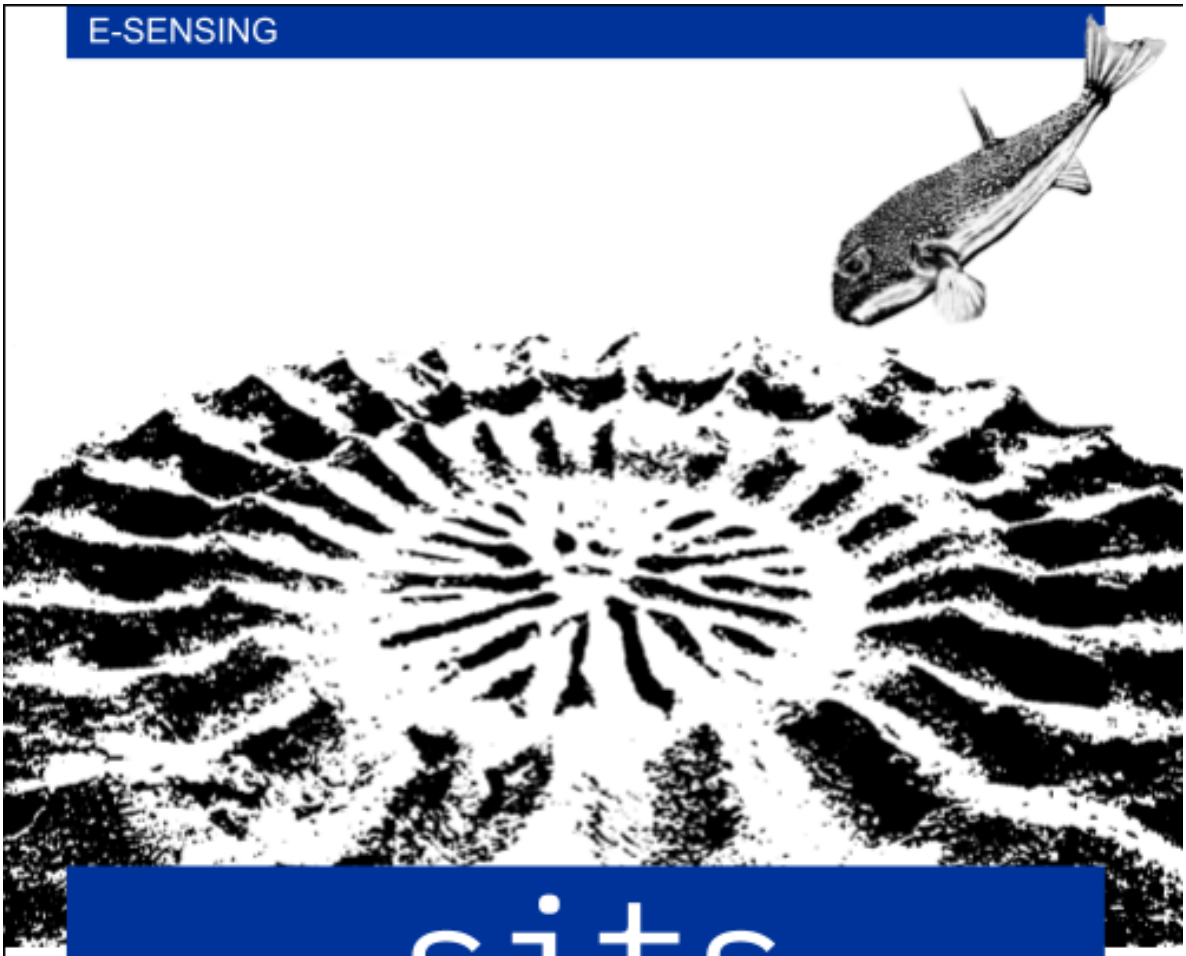
E-SENSING



sits

Satellite Image Time Series Analysis on Earth Observation Data Cubes

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Preface

Using time series derived from big Earth Observation data sets is one of the leading research trends in Land Use Science and Remote Sensing. One of the more promising uses of satellite time series is its application to classify land use and land cover. Information on land is critical for sustainable development because our growing demand for natural resources is causing significant environmental impacts. As stated by [1], *"time series analysis is expanding the kinds of land surface change that can be monitored using remote sensing. More subtle changes in ecosystem health and condition and related to land use dynamics are being monitored. The result is a paradigm shift away from change detection, typically using two points in time, to monitoring, or an attempt to track change continuously in time".*

This book presents `sits`, an open-source R package for land use and land cover classification using big Earth observation data. Users can build regular data cubes from collections in AWS, Microsoft Planetary Computer, Brazil Data Cube, and Digital Earth Africa. The software includes functions for quality assessment of training samples using self-organized maps. It provides machine learning and deep learning algorithms for classification of big Earth observation data cubes. Post-processing methods includes Bayesian smoothing and uncertainty assessment. Users can apply best practices for estimating area and assessing accuracy of land change. Thus, `sits` is an end-to-end toolkit for land mapping with Earth observation.

Who this book is for

The target audience for **sits** is the new generation of specialists who understand the principles of remote sensing and can write scripts in **R**. Ideally, users should have basic knowledge of data science methods using R. If you want more information on methods used in this book, please look at the following references:

- Garrett Golemund, "Hands-On Programming with R". O'Reilly, 2014. https://rstudio-education.github.io/hopr/.

- Hadley Wickham and Gareth Golemund, "R for Data Science". O'Reilly, 2017. https://r4ds.had.co.nz/.
- Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, "An Introduction to Statistical Learning with Applications in R", Springer, 2013. https://www.statlearning.com/.
- Zhang, A.; Zachary C. Lipton, Mu Li, Alexander J. Smola, "Dive into Deep Learning", 2021. https://d2l.ai/.

Main reference for sits

If you use sits in your work, please cite the following paper:

Rolf Simoes, Gilberto Camara, Gilberto Queiroz, Felipe Souza, Pedro R. Andrade, Lorena Santos, Alexandre Carvalho, and Karine Ferreira. "Satellite Image Time Series Analysis for Big Earth Observation Data". Remote Sensing, 13, p. 2428, 2021. https://doi.org/10.3390/rs13132428.

Setup

The `sits` package relies on the `sf` and `terra` R packages, which in turn require the GDAL and PROJ libraries. Please follow the instructions below for installing `sf` and `terra` together with GDAL, provided by Edzer Pebesma.

Support for GDAL and PROJ

Windows and MacOS

Windows and MacOS users are strongly encouraged to install the `sf` and `terra` binary packages from CRAN. To install `sits` from source, please install package `Rtools` to have access to the compiling environment.

Ubuntu

For Ubuntu versions 20.04 (focal) and 22.04 (jammy), we provide a script for fast installation, based on the work by Dirk Eddelbuettel in r2u[1] project. The script installs binary versions of sits and all its dependencies. It requires R version 4.2.0 or later. To download the script use:

```
wget https://raw.githubusercontent.com/e-sensing/sitsbook/master/utils/
    fast_sits_installation_ubuntu.sh
```

and run the script as sudo:

```
sudo sh fast_sits_installation_ubuntu.sh
```

Alternatively, the installation can be done step-by-step. We recommend using the latest version of the GDAL, GEOS, and PROJ4 libraries. To do so, use the repository `ubuntugis-unstable`, which should be done as follows:

```
sudo add-apt-repository ppa:ubuntugis/ubuntugis-unstable
sudo apt-get update
sudo apt-get install libudunits2-dev libgdal-dev libgeos-dev libproj-dev
```

[1] https://github.com/eddelbuettel/r2u

If you get an error while adding this PPA repository, it could be because you miss the package `software-properties-common`. When GDAL is running in `docker` containers, please add the security flag `--security-opt seccomp=unconfined` on start.

Debian

To install on Debian, use the rocker geospatial[2] dockerfiles.

Fedora

The following command installs all required dependencies:

```
sudo dnf install gdal-devel proj-devel geos-devel sqlite-devel udunits2-
    devel
```

Installing the `sits` package

`sits` is available on CRAN and should be installed as other R packages.

```
install.packages("sits")
```

The source code repository is on GitHub https://github.com/e-sensing/sits. To install the development version of `sits`, which contains the latest updates but might be unstable, users should install `devtools` if not already available, as then install sits as follows:

```
install.packages("devtools")
devtools::install_github("e-sensing/sits@dev", dependencies = TRUE)
```

To run the examples in the book, please also install the "sitsdata" package.

```
options(download.file.method = "wget")
devtools::install_github("e-sensing/sitsdata")
```

[2] https://github.com/rocker-org/geospatial

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1. Amazon Fund, established by the Brazil with financial contribution from Norway, through contract 17.2.0536.1. between the Brazilian Development Bank (BNDES) and the Foundation for Science, Technology and Space Applications (FUNCATE), for the establishment of the Brazil Data Cube,

2. Coordenação de Aperfeiçoamento de Pessoal de Nível Superior-Brasil (CAPES) and from the Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq) for grants 312151/2014-4 and 140684/2016-6.

3. Sao Paulo Research Foundation (FAPESP) under eScience Program grant 2014/08398-6, for for providing MSc, PhD and post-doc scholarships, equipment, and travel support.

4. International Climate Initiative of the Germany Federal Ministry for the Environment, Nature Conservation, Building and Nuclear Safety (IKI) under grant 17-III-084- Global-A-RESTORE+ ("RESTORE+: Addressing Landscape Restoration on Degraded Land in Indonesia and Brazil").

5. Microsoft Planetary Computer initiative under the GEO-Microsoft Cloud Computer Grants Programme.

Community Contributions

The authors thank Marius Appel, Tim Appelhans, Henrik Bengtsson, Robert Hijmans, Edzer Pebesma, and Ron Wehrens, for their R packages `gdalcubes`, `leafem`, `data.table`, `terra`, `sf/stars`, and "kohonen". We are indebted to the RStudio team, including Hadley Wickham for the `tidyverse` and Daniel Falbel for `torch`.

Reproducible papers used in building sits

We thank the authors of these papers for making their code available.

- Marius Appel and Edzer Pebesma, "On-Demand Processing of Data Cubes from Satellite Image Collections with the Gdalcubes Library." Data 4 (3): 1–16, 2020. https://doi.org/10.3390/data4030092

- Hassan Fawaz, Germain Forestier, Jonathan Weber, Lhassane Idoumghar, and Pierre-Alain Muller, "Deep learning for time series classification: a review". Data Mining and Knowledge Discovery, 33(4): 917–963, 2019. https://doi.org/10.1007/s10618-019-00619-1.

- Edzer Pebesma, "Simple Features for R: Standardized Support for Spatial Vector Data". R Journal, 10(1):2018.

- Charlotte Pelletier, Geoffrey Webb, and Francois Petitjean. "Temporal Convolutional Neural Network for the Classification of Satellite Image Time Series". Remote Sensing 11 (5), 2019. https://doi.org/10.3390/rs11050523

- Ron Wehrens and Kruisselbrink, Johannes. "Flexible Self-Organising Maps in kohonen 3.0". Journal of Statistical Software, 87, 7 (2018). https://doi.org/10.18637/jss.v087.i07.

- Vivien Garnot, Loic Landrieu, Sebastien Giordano, and Nesrine Chehata, "Satellite Image Time Series Classification with Pixel-Set Encoders and Temporal Self-Attention", Conference on Computer Vision and Pattern Recognition, 2020. https://doi.org/10.1109/CVPR42600.2020.01234.

- Vivien Garnot, Loic Landrieu, "Lightweight Temporal Self-Attention for Classifying Satellite Images Time Series", 2020. <arXiv:2007.00586>.

- Maja Schneider, Marco Körner, "[Re] Satellite Image Time Series Classification with Pixel-Set Encoders and Temporal Self-Attention." ReScience C 7 (2), 2021. doi:10.5281/zenodo.4835356.

Publications using sits

This section gathers the publications that have used **sits** to generate their results.

2021

- Lorena Santos, Karine R. Ferreira, Gilberto Camara, Michelle Picoli, Rolf Simoes, "Quality control and class noise reduction of satellite image time series". ISPRS Journal of Photogrammetry and Remote Sensing, vol. 177, pp 75-88, 2021. https://doi.org/10.1016/j.isprsjprs.2021.04.014.

- Lorena Santos, Karine Ferreira, Michelle Picoli, Gilberto Camara, Raul Zurita-Milla and Ellen-Wien Augustijn, "Identifying Spatiotemporal Patterns in Land Use and Cover Samples from Satellite Image Time Series". Remote Sensing, 2021, 13(5), 974; https://doi.org/10.3390/rs13050974.

2020

- Rolf Simoes, Michelle Picoli, Gilberto Camara, Adeline Maciel, Lorena Santos, Pedro Andrade, Alber Sánchez, Karine Ferreira & Alexandre Carvalho. "Land use and cover maps for Mato Grosso State in Brazil from 2001 to 2017". Nature Scientific Data 7, article 34 (2020). DOI: 10.1038/s41597-020-0371-4.

- Michelle Picoli, Ana Rorato, Pedro Leitão, Gilberto Camara, Adeline Maciel, Patrick Hostert, Ieda Sanches, "Impacts of Public and Private Sector Policies on Soybean and Pasture Expansion in Mato Grosso—Brazil from 2001 to 2017". Land, 9(1), 2020. DOI: 10.3390/land9010020.

- Karine Ferreira, Gilberto Queiroz et al., "Earth Observation Data Cubes for Brazil: Requirements, Methodology and Products". Remote Sensing, 12, 4033, 2020.

- Adeline Maciel, Lubia Vinhas, Michelle Picoli and Gilberto Camara, "Identifying Land Use Change Trajectories in Brazil's Agricultural Frontier". Land, 9, 506, 2020. DOI: 10.3390/land9120506. DOI: 10.3390/rs12244033.

2018

- Michelle Picoli, Gilberto Camara, et al., "Big Earth Observation Time Series Analysis for Monitoring Brazilian Agriculture". ISPRS Journal of Photogrammetry and Remote Sensing, 2018.

Introduction to SITS

How `sits` works

The `sits` package is based on the premise of using all of the data available in an Earth observation data cube, adopting a *time-first, space-later* approach. Each spatial location is associated to a time series. Locations with known labels train a machine learning classifier, which classifies all time series of a data cube, as shown in Figure 1.

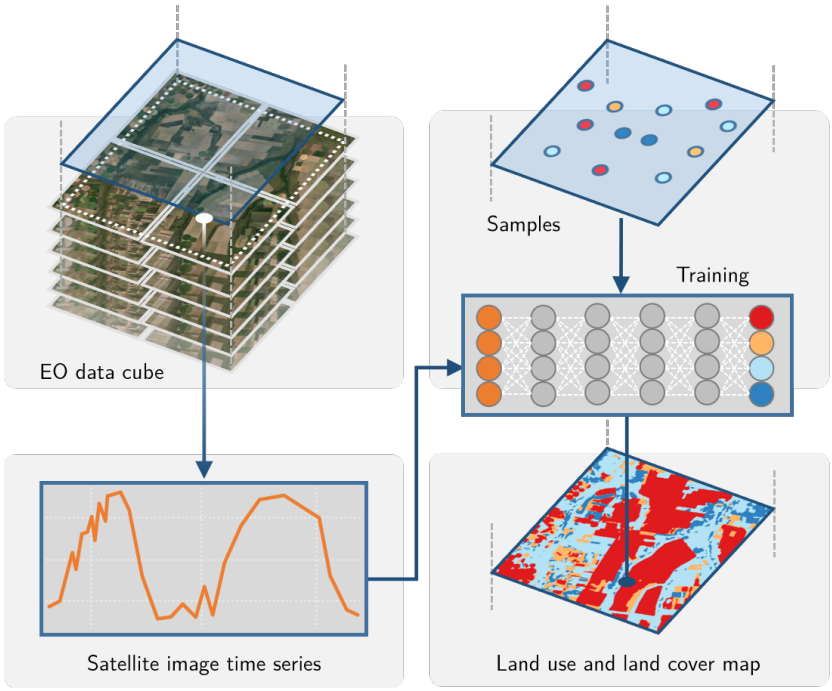

Figure 1: Using time series for land classification (source: authors)